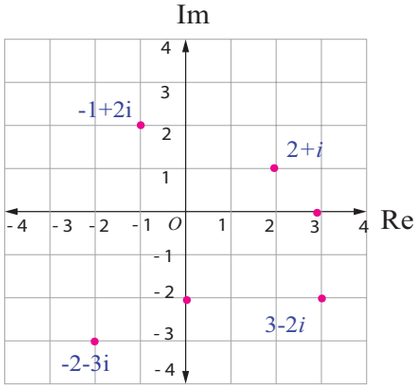
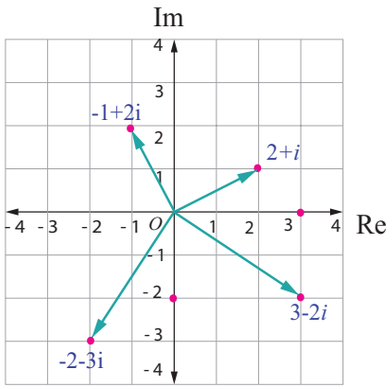


Complex numbers as points

Fig. 2.6



Complex numbers as vectors

Fig. 2.7

2.2.3 Algebraic operations on complex numbers

In this section, we study the algebraic and geometric structure of the complex number system. We assume various corresponding properties of real numbers to be known.

(i) Scalar multiplication of complex numbers:

If $z = x + iy$ and $k \in \mathbb{R}$, then we define

$k\,z = (kx) + (ky)\,i$.

In particular $0\,z = 0$, $1\,z = z$ and $(-1)z = -z$.

The diagram below shows $k\,z$ for $k = 2, \frac{1}{2}, -1$

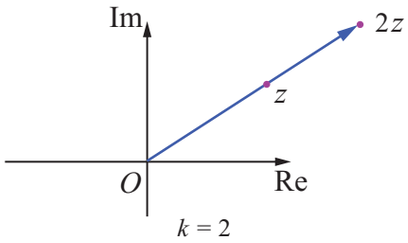


Fig. 2.8

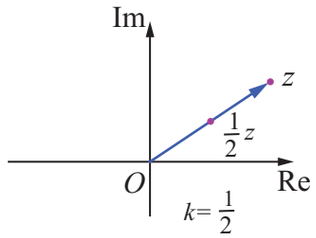


Fig. 2.9

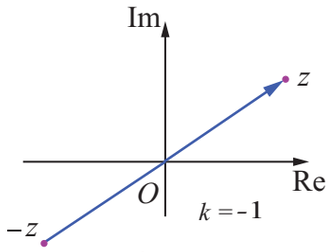


Fig. 2.10

(ii) Addition of complex numbers:

If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, where $x_1, x_2, y_1,$ and $y_2 \in \mathbb{R}$, then we define

$$\begin{aligned} z_1 + z_2 &= (x_1 + iy_1) + (x_2 + iy_2) \\ &= (x_1 + x_2) + i(y_1 + y_2) \\ z_1 + z_2 &= (x_1 + x_2) + i(y_1 + y_2). \end{aligned}$$

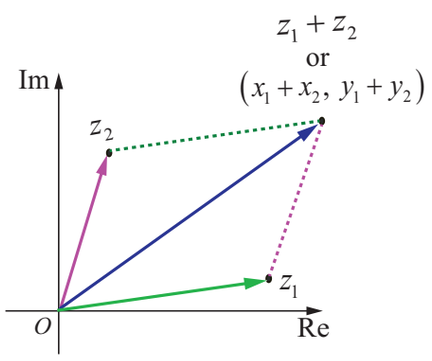


Fig. 2.11

We have already seen that vectors are characterized by length and direction, and that a given vector remains unchanged under translation. When $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ then by the parallelogram law of addition, the sum $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$ corresponds to the point $(x_1 + x_2,\ y_1 + y_2)$. It also corresponds to a vector with those coordinates as its components. Hence the points z_1, z_2, and $z_1 + z_2$ in complex plane may be obtained vectorially as shown in the adjacent Fig. 2.11.

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